

The code for `$a_2$` is:

Please provide a brief explanation of your answer.

Explanation:

- The code defines a_2 as the limit of $2f(n) + \frac{1}{2} + 0.25^n$ as $n \rightarrow \infty$.
- $f(n)$ sums $\frac{1}{2^k}$ terms (converging to $\frac{1}{4}$) plus a tiny positive amount from the 72^{-A} terms (where $A = 10^{24}$).
- Thus $a_2 = 1 + \delta$ for some extremely small $\delta > 0$. Since $a_2 > 1$, the optimal solution to minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ is $x_1 = 0$, $x_2 = \frac{1}{a_2} \approx 1$.
- Distance in the infinity norm from $(0.3, 0.7)$ to $(0, \approx 1)$ is $\max(0.3, \approx 0.3) = 0.3 \leq 0.4 \rightarrow \text{True}$.
- Distance from $(0.6, 0.4)$ to $(0, \approx 1)$ is $\max(0.6, \approx 0.6) = 0.6 > 0.45 \rightarrow \text{False}$.